

Dylan R. Zelko

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EDUCATION

Master of Science - Robotics <i>University at Buffalo, School of Engineering and Applied Sciences</i>	Expected May 2027 GPA: 3.5
Bachelor of Science - Mechanical Engineering; Minor in Robotics <i>University at Buffalo, School of Engineering and Applied Sciences</i>	Sept. 2021 - May 2025

RELEVANT WORK EXPERIENCE

Digital Manufacturing Lab, University at Buffalo	August 2025 - Present
- Fabricate and manage 3D printing work orders using FDM, SLA, PolyJet, and CFF technologies, ensuring efficient turnaround times with high-quality prints - Provide technical support and guidance to students, faculty, and industry on best practices for 3D printing, including material selection, print orientation, support strategies, and post-processing techniques	
Graduate Teaching Assistant MAE 451/494, University at Buffalo	August 2025 - Present
- Provide detailed feedback on senior design projects, including mechanical designs, CAD models, and technical documentation - Collaborate with faculty to ensure consistent evaluation of project milestones and deliverables across multiple design teams	

PROJECTS

NASA Lunabotics Competition, University at Buffalo <i>Founder & Project Lead, University at Buffalo</i>	Sept. 2024 - Present Link to Demo
- Spearheaded the mechanical CAD design and fabrication of a novel lunar rover, including custom wheels to improve traction and durability on simulated lunar terrain, a modular chassis, and a rotating drum excavator - Conducted FEA simulations using ANSYS to optimize structural components - Developed an efficient regolith collection system using Discrete Element Methods (DEM) to optimize geometry and operating conditions - Awarded best first-year's presentation and demonstration at the NASA Lunabotics Competition	
Mars Rover Research Team, University at Buffalo <i>Founder & Project Lead, University at Buffalo</i>	April 2023 - May 2025 Link to Demo
- Founded a 50-member multidisciplinary research team to design next-generation rovers for the University Rover Challenge - Coordinated cross-functional teams in mechanical, electrical, and software engineering to develop robust robotic systems for extreme environments - Prepared technical documentation, including project management plans and systems engineering report	

SKILLS

Technical Skills: Fusion 360, SolidWorks, ANSYS, Microsoft Suite (Excel, Teams, PowerPoint), 3D printing (FDM, SLA, PolyJet, CFF) including printer maintenance and material selection, machining

Programming: MATLAB, Python, ROS2, C ++

SCHOLARSHIPS, AWARDS, and ACHIEVEMENTS

- NASA New York Space Grant Internship Award, 2025
- Barbara McDaniel Undergraduate Scholarship by the National Training and Simulation Association, 2024
- President, UB Robotics Club, Sept. 2022 - May 2025
- Publication: K. Wang, D. Zelko, M. Delano, *Textile band electrodes as an alternative to spot Ag/AgCl electrodes for calf bioimpedance measurements*, Biomedical Physics & Engineering Express, 6(1), 015010, 2020. [doi:10.1088/2057-1976/ab5b02](https://doi.org/10.1088/2057-1976/ab5b02)